

Tanzim Islam Khan
dihan2468@gmail.com
May 25, 2026

May 25, 2026

The Editor-in-Chief
Information Technology for Development
Taylor & Francis

Dear Editor-in-Chief,

I am pleased to submit for your consideration the manuscript titled:

Generative AI Adoption in the Bangladeshi Software Engineering Workforce: A Simulation-Based Analysis of Diffusion, Productivity, and Skills Gaps

What the paper does. This paper presents the first integrated quantitative account of generative AI (GenAI) adoption in the Bangladeshi software engineering workforce between 2022 and 2026. The analysis combines a Bass diffusion model calibrated to the adoption window, a reproducible synthetic workforce simulation ($N = 1,500$) anchored to BASIS, BBS, and World Bank population statistics, and a battery of statistical analyses including heterogeneous treatment effect estimation by English proficiency stratum and a skills gap index mapped against university tier. Four policy scenarios are projected to 2031.

Why it fits *Information Technology for Development*. The journal's stated scope covers theoretical, empirical, and critical research on the role of information technology in social, economic, and human development outcomes in developing countries. This manuscript addresses that scope directly on four dimensions:

- *Country focus:* the analysis is specific to Bangladesh, a large lower-middle-income country in which the IT sector is a primary vehicle for export-led growth and employment.
- *Development relevance:* the findings are framed explicitly around SDG 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation and Infrastructure), both of which are central to Bangladesh's Vision 2041 development agenda.
- *LMIC-specific insight:* the paper identifies English language proficiency as a robust structural moderator of GenAI productivity effects — a finding specific to LMIC contexts that the dominant North American and European literature cannot supply.
- *Policy utility:* the projected workforce scenarios and the skills gap index provide a quantitative basis for national digital literacy and university curriculum policy, directly usable by a2i, the ICT Division, and the University Grants Commission of Bangladesh.

Originality. To the best of my knowledge, no prior study integrates a diffusion model, a calibrated workforce simulation, and a policy projection specific to Bangladesh. The paper fills this gap and provides a replicable analytical framework that can be adapted to other comparable LMIC contexts (Vietnam, Philippines, Nigeria, Egypt).

Reproducibility. All source code (Python), 40 automated unit tests, 12 publication-quality figures, and 11 data tables are released under open licences at:

<https://github.com/tanzimkhan24/genai-se-bangladesh>

Submission declaration. This manuscript has not been published elsewhere and is not currently under review by any other journal or conference. All work is original. No external funding was received for this research. The author declares no conflicts of interest. This study uses simulated data calibrated to publicly reported aggregate statistics; no human subjects research was conducted and no ethics approval was required.

Suggested reviewers. I suggest the following scholars as potential reviewers, given their expertise in ICT4D, technology adoption in LMICs, and the Bangladeshi IT sector:

1. A researcher specialising in Bass diffusion modelling in emerging-economy ICT contexts.
2. A researcher specialising in GenAI and software engineering productivity in South or Southeast Asia.
3. A researcher specialising in computing education and skills gaps in LMIC university systems.

I am happy to provide specific names on request, or to defer entirely to the Editor's discretion.

I look forward to receiving the reviewers' comments and to the possibility of publishing this work in *Information Technology for Development*.

Yours sincerely,

Tanzim Islam Khan